Lazizbek Sadullaev. Math 464. HW #4.

For the given problem, we need to optimize the number of trees to be planted that satisfy the constraints given in the problem.

Let x_1 be the number of apples
x_2 be the number of pears
x_3 be the number of cherries.

So the optimization problem is as follows:

$$\max \quad z = x_1 + x_2 + x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3$$

s.t.

$$x_1 + x_2 + x_3 \leq 345 \rightarrow \text{max. \# of trees}$$

$$150\, x_1 + 200\, x_2 + 240\, x_3 \leq 50.000 \rightarrow \text{Labor cost constrain}$$

$$275\, x_1 + 180\, x_2 + 125\, x_3 \leq 60.000 \rightarrow \text{material cost constrain}$$

(1)

$$150\, y_1 \leq x_1 \leq 345\, y_1 \quad \text{where } y_1 \in \{0, 1\}$$

$$50 \cdot y_2 \leq x_2 \leq 345\, y_2 \quad \text{where } y_2 \in \{0, 1\}$$

$$80\, y_3 \leq x_3 \leq 345\, y_3 \quad \text{where } y_3 \in \{0, 1\}$$

$$x = [x_1, x_2, x_3]^T \geq 0$$

$$x \in \mathbb{Z}^3.$$

Now, let me try to adjust this maximization problem so that program (software) can understand:

We have new (additional) decision variables y_1, y_2, y_3 to satisfy the problem requirements. But these variables (y_1, y_2, y_3) are not participating in the objective function $z = x_1 + x_2 + x_3$, so I added them with <u>zero</u> coefficients to make them appear in z. Now, (i) we should do the same thing to constrain inequalities, and (ii) write the last inequalities in the standard format. Let's do these 2 steps:

(i) $150x_1 + 200x_2 + 240x_3 \leq 50.000$ → Labor constrain

$\Downarrow$

(2) $150x_1 + 200x_2 + 240x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3 \leq 50.000$

similarly

$275x_1 + 180x_2 + 125x_3 \leq 60.000$ → material constrain

$\Downarrow$

(3) $275x_1 + 180x_2 + 125x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3 \leq 60.000$

(ii) For the last 3 constrains in (1) where we used binary desicion variables, let's write them in the standard form similar to (2) & (3):

$$150 y_1 \le x_1 \le 345 y_1 \quad \begin{cases} 150 y_1 \le x_1 & (4) \\ x_1 \le 345 y_1 & (5) \end{cases}$$

(4) can be written as:

$150 y_1 - x_1 \le 0$ & making it standard

(6) $-x_1 + 0 \cdot x_2 + 0 \cdot x_3 + 150 y_1 + 0 \cdot y_2 + 0 \cdot y_3 \le 0$

Similarly, (5) can be written as:

$x_1 - 345 y_1 \le 0$ & making is standard:

(7) $x_1 + 0 x_2 + 0 \cdot x_3 - 345 y_1 + 0 \cdot y_2 + 0 \cdot y_3 \le 0$

So the following substitution is valid:

$$150 y_1 \le x_1 \le 345 y_1 \iff \begin{cases} (6)\ -x_1 + 0 \cdot x_2 + 0 \cdot x_3 + 150 y_1 + 0 \cdot y_2 + 0 \cdot y_3 \le 0 \\ \& \\ (7)\ x_1 + 0 \cdot x_2 + 0 \cdot x_3 - 345 y_1 + 0 \cdot y_2 + 0 \cdot y_3 \le 0 \end{cases}$$

(8) where $0 \le y_1 \le 1$ & $y_1 \in \mathbb{Z}$

We can write the remaing 2 constraints in a similar fashion.

$$50 y_2 \leq x_2 \leq 345 y_2 \qquad y_2 \in \{0,1\} \iff$$

$$\iff \quad 0 \cdot x_1 - x_2 + 0 \cdot x_3 + 0 \cdot y_1 + 50 y_2 + 0 \cdot y_3 \leq 0$$

$$(9) \quad 0 \cdot x_1 + x_2 + 0 \cdot x_3 + 0 \cdot y_1 - 345 y_2 + 0 \cdot y_3 \leq 0$$

where $0 \leq y_2 \leq 1 \qquad y_2 \in \mathbb{Z}$,

and finally

$$80 y_3 \leq x_3 \leq 345 y_3 \quad , \quad y_3 \in \{0,1\} \iff$$

$$0 \cdot x_1 + 0 \cdot x_2 - x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 80 y_3 \leq 0$$

$$\iff$$

$$(10) \quad 0 \cdot x_1 + 0 \cdot x_2 + x_3 + 0 \cdot y_1 + 0 \cdot y_2 - 345 y_3 \leq 0$$

where $0 \leq y_3 \leq 1 \quad , \quad y_3 \in \mathbb{Z}$.

By replacing these 3 constraints by their respective equivalent constraints the maximal optimization problem (1) becomes

$$\max_{x,y} \quad z = x_1 + x_2 + x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3$$

s.t.

$$
\begin{aligned}
x_1 + x_2 + x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3 &\leq 345 \\
150 x_1 + 200 x_2 + 240 x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3 &\leq 50.000 \\
275 x_1 + 180 x_2 + 125 x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3 &\leq 60.000 \\
-x_1 + 0 \cdot x_2 + 0 \cdot x_3 + 150 y_1 + 0 \cdot y_2 + 0 \cdot y_3 &\leq 0 \\
x_1 + 0 \cdot x_2 + 0 \cdot x_3 - 345 y_1 + 0 \cdot y_2 + 0 \cdot y_3 &\leq 0 \\
0 \cdot x_1 - x_2 + 0 \cdot x_3 + 0 \cdot y_1 + 50 y_2 + 0 \cdot y_3 &\leq 0 \\
0 \cdot x_1 + x_2 + 0 \cdot x_3 + 0 \cdot y_1 - 345 y_2 + 0 \cdot y_3 &\leq 0 \\
0 \cdot x_1 + 0 \cdot x_2 - x_3 + 0 \cdot y_1 + 0 \cdot y_2 + 80 y_3 &\leq 0 \\
0 \cdot x_1 + 0 \cdot x_2 + x_3 + 0 \cdot y_1 + 0 \cdot y_2 - 345 y_3 &\leq 0
\end{aligned}
\tag{11}
$$

$$x \geq 0$$

$$0 \leq y \leq 1$$

$$x \in \mathbb{Z}^3, \qquad y \in \mathbb{Z}^3$$

I'll be solving this (11) optim. problem by software and attaching the results separately:

$$x^* = [\overset{x_1}{172}, \overset{x_2}{0}, \overset{x_3}{100}, \overset{y_1}{1}, \overset{y_2}{0}, \overset{y_3}{1}]^T \quad \&$$

$$z^* = 172 + 100 = 272.$$

math_464_actual_hw4

February 6, 2024

Math 464. HW4. Lazizbek Sadullaev

1 *Integer Program*

[9]:
```
"""
Script to demonstrate how to solve small mixed integer programs
using the python optimize module in the scipy package
"""
"""
All decision variables are from reals:
"""

import pandas as pd
import numpy as np
import scipy.optimize as opt

# The problem we will solve is:
#
#    max z =  x1 + x2 +  x3 + 0 y1 + 0 y2 + 0 y3
#   s.t.      x1 +     x2 +     x3 + 0 y1 + 0 y2 + 0 y3 <= 345
#         150 x1 + 200 x2 + 240 x3 + 0 y1 + 0 y2 + 0 y3 <= 50 000 - labor cost␣
 ↪constraints
#         275 x1 + 180 x2 + 125 x3 + 0 y1 + 0 y2 + 0 y3 <= 60 000 - material␣
 ↪cost constraints
#
#         -1 x1 + 0 x2 + 0 x3 + 150 y1 +   0 y2 +   0 y3 <= 0
#          1 x1 + 0 x2 + 0 x3 - 345 y1 +   0 y2 +   0 y3 <= 0     # 150 y1 <=␣
 ↪x1 <= 345 y1
#          0 x1 - 1 x2 + 0 x3 +   0 y1 +  50 y2 +   0 y3 <= 0
#          0 x1 + 1 x2 + 0 x3 +   0 y1 - 345 y2 +   0 y3 <= 0     # 50 y2 <= x2␣
 ↪<= 345 y2
#          0 x1 + 0 x2 - 1 x3 +   0 y1 +   0 y2 +  80 y3 <= 0
#          0 x1 + 0 x2 + 1 x3 +   0 y1 +   0 y2 - 345 y3 <= 0     # 80 y3 <= x3␣
 ↪<= 345 y3
#
#               0 <= y1 <= 1
#               0 <= y2 <= 1
```

```
#               0 <= y3 <= 1
#               x >= 0
#               y >= 0
#            x1, x2,x3 in Z
#            y1, y2,y3 in Z


# First build the objective vector.
c=np.array([1, 1, 1, 0, 0, 0])

# Next, create the coefficient array for the inequality constraints.
# Note that the inequalities must be Ax <= b, so some sign
# changes result when converting >= into <=.
A = np.array([[   1,   1,   1,    0,    0,    0],\
              [ 150, 200, 240,    0,    0,    0],\
              [ 275, 180, 125,    0,    0,    0],\
              [  -1,   0,   0,  150,    0,    0],\
              [   1,   0,   0, -345,    0,    0],\
              [   0,  -1,   0,    0,   50,    0],\
              [   0,   1,   0,    0, -345,    0],\
              [   0,   0,  -1,    0,    0,   80],\
              [   0,   0,   1,    0,    0, -345]])

# Next the right-hand-side vector for the inequalities
# Sign changes can occur here too.
b = np.array([345 , 50000 , 60000, 0, 0, 0, 0, 0, 0 ])

#The coefficient matrix for the equality constraints and
# the right hand side vector.
Ae = None     # Ae = [[1,1,1,1]]
be = None

# Next, we provide any lower and upper bound vectors, one
# value for each decision variable.  In this example all
# lower bound are zero and there are no upper bounds.
bounds=((0,np.inf),(0,np.inf),(0,np.inf),(0,1), (0,1), (0,1))

# Lastly, we can specify which variables are required to be integer.
# If no variables are integer then isint=[];  In our example, only x2
# is integer.
isint=[1, 1, 1, 1, 1, 1]

# The call to the mixed integer solver looks like the following.
# Notice that we pass usual "c" when we have a minimization
# problem, we send "-c" when we have max problem.
# This is because the solver is expecting a minimization.
```

```
res=opt.linprog(-c,A,b,Ae,be,bounds,integrality=isint)

# The result is stored in the dictionary variable "res".
# In particular, to show the optimal objective value and the
# optimal decision variable values:
print("min z = ", np.dot(res['x'][:6], A[0][:6]))
print(res['fun'])
print(res['x'])
print(np.dot(res['x'], A[0])) # to verify if the answer is corret
print(np.dot(res['x'], A[1]))
print(np.dot(res['x'], A[2]))

# print(res)
```

```
min z =  272.0
-272.0
[172.   0. 100.   1.   0.   1.]
272.0
49800.0
59800.0
```

[ ]:
```
!sudo apt-get install texlive-xetex texlive-fonts-recommended␣
↪texlive-plain-generic
```

```
Reading package lists… Done
Building dependency tree… Done
Reading state information… Done
The following additional packages will be installed:
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
  fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
  libcommons-parent-java libfontbox-java libfontenc1 libgs9 libgs9-common
  libidn12 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1
  libruby3.0 libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1
  libzzip-0-13 lmodern poppler-data preview-latex-style rake ruby
  ruby-net-telnet ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0
  rubygems-integration t1utils teckit tex-common tex-gyre texlive-base
  texlive-binaries texlive-latex-base texlive-latex-extra
  texlive-latex-recommended texlive-pictures tipa xfonts-encodings
  xfonts-utils
Suggested packages:
  fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
  libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
  poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
  fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
  fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
  | postscript-viewer perl-tk xpdf | pdf-viewer xzdec
  texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
  icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
```

```
  texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
  texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
  default-jre-headless tipa-doc
The following NEW packages will be installed:
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
  fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
  libcommons-parent-java libfontbox-java libfontenc1 libgs9 libgs9-common
  libidn12 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1
  libruby3.0 libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1
  libzzip-0-13 lmodern poppler-data preview-latex-style rake ruby
  ruby-net-telnet ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0
  rubygems-integration t1utils teckit tex-common tex-gyre texlive-base
  texlive-binaries texlive-fonts-recommended texlive-latex-base
  texlive-latex-extra texlive-latex-recommended texlive-pictures
  texlive-plain-generic texlive-xetex tipa xfonts-encodings xfonts-utils
0 upgraded, 54 newly installed, 0 to remove and 32 not upgraded.
Need to get 182 MB of archives.
After this operation, 571 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-droid-fallback all
1:6.0.1r16-1.1build1 [1,805 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-lato all 2.0-2.1
[2,696 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 poppler-data all
0.4.11-1 [2,171 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-common all 6.17
[33.7 kB]
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20200910-1 [6,367 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9-common
all 9.55.0~dfsg1-0ubuntu5.6 [751 kB]
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1.38-4ubuntu1 [60.0 kB]
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9.55.0~dfsg1-0ubuntu5.6 [5,031 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libkpathsea6
amd64 2021.20210626.59705-1ubuntu0.1 [60.3 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy/main amd64 libwoff1 amd64
1.0.2-1build4 [45.2 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy/universe amd64 dvisvgm amd64
2.13.1-1 [1,221 kB]
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2.004.5-6.1 [4,532 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-noto-mono all
20201225-1build1 [397 kB]
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20180621-3.1 [10.2 MB]
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all 18-1 [4,720 B]
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java all 43-1 [10.8 kB]
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java all 1.2-2 [60.3 kB]
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1:1.1.4-1build3 [14.7 kB]
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3.3.5-2 [228 kB]
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[5,100 B]
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0.13.72+dfsg.1-1.1 [27.0 kB]
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1:1.0.5-0ubuntu2 [578 kB]
Get:37 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-utils amd64
1:7.7+6build2 [94.6 kB]
Get:38 http://archive.ubuntu.com/ubuntu jammy/universe amd64 lmodern all
2.004.5-6.1 [9,471 kB]
Get:39 http://archive.ubuntu.com/ubuntu jammy/universe amd64 preview-latex-style
all 12.2-1ubuntu1 [185 kB]
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1.41-4build2 [61.3 kB]
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2.5.11+ds1-1 [699 kB]
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20180621-3.1 [6,209 kB]
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binaries amd64 2021.20210626.59705-1ubuntu0.1 [9,848 kB]
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generic all 2021.20220204-1 [27.5 MB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 15s (12.0 MB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based
frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 78,
<> line 54.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (This frontend requires a controlling tty.)
debconf: falling back to frontend: Teletype
dpkg-preconfigure: unable to re-open stdin:
Selecting previously unselected package fonts-droid-fallback.
(Reading database … 121747 files and directories currently installed.)
Preparing to unpack …/00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
…
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) …
Selecting previously unselected package fonts-lato.
Preparing to unpack …/01-fonts-lato_2.0-2.1_all.deb …
Unpacking fonts-lato (2.0-2.1) …
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Selecting previously unselected package poppler-data.
Preparing to unpack …/02-poppler-data_0.4.11-1_all.deb …
Unpacking poppler-data (0.4.11-1) …
Selecting previously unselected package tex-common.
Preparing to unpack …/03-tex-common_6.17_all.deb …
Unpacking tex-common (6.17) …
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack …/04-fonts-urw-base35_20200910-1_all.deb …
Unpacking fonts-urw-base35 (20200910-1) …
Selecting previously unselected package libgs9-common.
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Selecting previously unselected package libijs-0.35:amd64.
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Unpacking libijs-0.35:amd64 (0.35-15build2) …
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack …/08-libjbig2dec0_0.19-3build2_amd64.deb …
Unpacking libjbig2dec0:amd64 (0.19-3build2) …
Selecting previously unselected package libgs9:amd64.
Preparing to unpack …/09-libgs9_9.55.0~dfsg1-0ubuntu5.6_amd64.deb …
```

[ ]:
```
!jupyter nbconvert --to pdf /content/math_464_actual_hw2.ipynb
```

```
[NbConvertApp] Converting notebook /content/math_464_actual_hw2.ipynb to pdf
[NbConvertApp] Writing 36815 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | bibtex had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 51449 bytes to /content/math_464_actual_hw2.pdf
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